



Active but Hasty? CEO age, ownership structure, and the quantity–quality trade-off in investment[☆]

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ABSTRACT

We examine whether the well-documented tendency of young CEOs to invest more aggressively translates into efficient capital allocation. Estimating both the quantity (capex sensitivity to sales growth) and quality (return on invested capital) of investment across CEO age groups within a unified framework, we find that young CEOs (aged 39 and under) exhibit the highest investment sensitivity but the lowest efficiency—an “Active but Hasty” pattern. The pattern varies systematically with ownership structure: the quantity–quality gap is most acute in solo-owner firms, mitigated in family-owner firms—consistent with kinship-based monitoring—and driven by an efficiency deficit in non-owner firms. These findings suggest that the value of youth-driven investment activity depends critically on the governance environment in which it occurs.

1. Introduction

Upper echelons theory predicts that managerial characteristics shape corporate decisions (Hambrick and Mason, 1984), and a robust body of empirical work confirms that CEO age is among the most consequential of these characteristics for investment. Younger CEOs spend more on R&D (Serfling, 2014), pursue more acquisitions (Yim, 2013), and invest more in physical capital (Belenzon et al., 2019). This “active youth” pattern carries implications for aging economies, where a rising share of firms is led by older executives. Yet whether such aggressiveness translates into *efficient* capital allocation — the quality, not merely the quantity, of investment — has received far less scrutiny.

The existing literature, however, conflates investment *quantity* with investment *quality*. Deviations from optimal investment — particularly overinvestment driven by agency costs — destroy value (Richardson, 2006), yet whether the well-documented aggressiveness of young CEOs represents efficient allocation or merely higher volumes of poorly targeted spending remains open. Pan et al. (2016) document that investment quantity rises and quality declines over CEO *tenure* as entrenchment progresses, but the analogous question for CEO *age* — which reflects career-stage incentives distinct from accumulated positional power — remains largely unexamined.

We address this gap by estimating both dimensions of investment behavior across CEO age groups within a unified framework: the *quantity* of investment, measured as the sensitivity of capital expenditure to lagged sales growth, and the *quality* of investment,

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measured as the return on invested capital (ROIC) conditional on the same growth signal. Both margins are estimated under the same firm-by-CEO and industry-by-year fixed effects and the same age-group interaction structure, placing quantity and quality on a directly comparable footing. This joint framework allows us to distinguish CEOs who invest both aggressively and efficiently from those who are merely aggressive.

Our setting — a large panel of listed and unlisted Japanese firms from Tokyo Shoko Research (TSR) over 1998–2019 — makes this feasible in ways public-firm data cannot: firm-by-CEO fixed effects absorb match-level selection, and granular records linking CEO names to both shareholder lists and executive rosters enable a decomposition into solo-owner, family-owner, and non-owner firms at a scale rarely available elsewhere.

Our main findings are threefold. First, confirming prior work, young CEOs (aged 39 and under) exhibit the highest investment sensitivity to sales growth. The sensitivity declines with CEO age, even after controlling for firm age, financial conditions, CEO tenure, and absorbing firm-by-CEO fixed effects.

Second, and more novel, young CEOs exhibit the *lowest* investment efficiency: their ROIC response to the same growth signal is flat or negative, while CEOs in their 50s and 60s generate significantly higher returns. We label this divergence the “Active but Hasty” pattern.

Third, the severity and form of this divergence vary across ownership-based governance environments: the quantity–quality gap is widest in solo-owner firms, mitigated in family-owner firms, and, among non-owner firms, takes the form of an efficiency deficit rather than excess investment—patterns consistent with unchecked discretion, kinship-based monitoring, and career concerns, respectively.

We contribute in two ways. The first is methodological: rather than examining either margin in isolation, we estimate the quantity and quality of investment within a single unified framework, introducing this joint decomposition to the CEO age–investment literature. This design is what reveals that age-related differences in investment *levels* do not map onto differences in *efficiency*—a result without direct precedent. Second, we provide micro-level evidence on the governance environments in which youthful aggressiveness creates or destroys value, linking the CEO age literature to work on ownership structure and investment efficiency.

2. Theoretical framework and hypothesis development

Theory yields competing predictions for how CEO age maps into investment *quantity* and *quality*, and the two need not coincide.

For young CEOs, even the sign of the investment effect is contested. Career-concern models in which unproven managers protect fragile reputations predict conservatism and herding (Holmström, 1999; Scharfstein and Stein, 1990), whereas an opposing tradition holds that young managers overweight private signals and exaggerate their investment decisions to build a reputation for ability (Prendergast and Stole, 1996), reinforced by compensation incentives that reward early deal-making (Yim, 2013) and by the greater risk tolerance of younger executives (Hambrick and Mason, 1984). For older CEOs, the horizon problem predicts retrenchment as retirement nears — documented empirically as cuts to long-horizon spending such as R&D (Dechow and Sloan, 1991; Serfling, 2014) — while a competing legacy channel holds that reputational or empire-building motives, freed from market discipline late in a career, can instead induce overinvestment of free cash flow where oversight is weak (Jensen, 1986; Stulz, 1990). Because these forces carry opposite signs for older CEOs, the age profile of investment sensitivity is itself a test of which channel dominates. Based on these competing predictions, we derive the following hypotheses:

H1 (Quantity). The sensitivity of investment to growth opportunities is highest for the youngest CEOs and lower among older age groups.

Whether such activity is efficient is a further question—one that the quantity channel alone cannot answer. Accumulated experience, financial expertise, and social capital sharpen project screening (Custódio and Metzger, 2014), implying quality that rises with age; however, two forces attenuate this premium at the oldest ages. First, Hambrick and Mason (1984) argue that cognitive flexibility and openness to new information decline with age, impairing the quality of strategic judgment. Second, the horizon-driven disengagement that curtails late-career investment quantity may also erode the attention devoted to project screening. The joint operation of these forces suggests an inverted-U profile: investment quality rises through the prime years and attenuates at the end of the career.

H2 (Quality). Young CEOs convert growth opportunities into returns least effectively; investment quality peaks among prime-age CEOs and attenuates at the oldest ages.

Jointly, H1 and H2 define a divergence between investment quantity and quality, whose value consequences are shaped in part by the governance environment. Ownership type proxies for three distinct governance environments that differ in monitoring intensity, incentive alignment, and managerial discretion. In solo-owner firms, dominant insiders combine concentrated control with limited board oversight. Equity ownership aligns incentives with firm value (Jensen and Meckling, 1976), but where internal oversight is absent, the same discretion that enables decisive investment also leaves expansionary impulses unchecked, generating the agency costs of free cash flow (Jensen, 1986; Stulz, 1990; Morck et al., 1988). In family-owner firms, kinship-based monitoring and a concern for transgenerational continuity discipline speculative investment while preserving initiative—a monitoring-driven setting (Anderson and Reeb, 2003; Gómez-Mejía et al., 2007). In non-owner firms, professional managers lacking equity face the weakest incentive alignment. Career concerns operate differently here than in the young-CEO context: without equity, the horizon for personal gain shortens to observable short-term outcomes, so the pressure to signal competence distorts project selection toward visible results over long-run returns (Gibbons and Murphy, 1992), rather than inducing the conservatism that career concerns produce when managerial ability is the primary unknown (Holmström, 1999).

H3 (Governance environments). The youth-related quantity–quality gap is widest in solo-owner firms (managerial discretion); attenuated in family-owner firms (monitoring intensity); and, in non-owner firms, takes the form of an efficiency deficit driven by career concerns rather than excess investment (weak incentive alignment).

3. Data and methodology

3.1. Data

We draw on two proprietary datasets compiled by Tokyo Shoko Research (TSR), one of the largest credit reporting agencies in Japan. The Corporate Information File (CI File) provides firm and CEO attributes including industry, location, founding year, board composition, major shareholders, CEO name, and CEO birth date. The Financial Statement File (KESSAN File) provides balance sheet and income statement items including tangible fixed assets, land, depreciation, operating income, and total liabilities. The sample spans fiscal years 1998–2019 and encompasses both listed and unlisted firms. Our analysis uses the subset of firms for which both CEO attributes and financial statement data are available; this merged panel grows from approximately 56,000 firms in 1998 to over 300,000 in the mid-2010s.¹ Summary statistics for both samples are reported in Table A1 in the Online Appendix.

We construct the capital expenditure ratio as the year-over-year change in tangible fixed assets net of land plus depreciation, divided by the lagged stock, restricting the sample to firm-years with a positive lagged stock to ensure a well-defined denominator. After removing missing values and trimming the top and bottom 1% of the capex ratio, lagged sales growth, log employees, leverage ratio, and EBITDA ratio, the investment quantity sample comprises approximately 2.5 million firm-year observations. The ROIC sample is smaller (approximately 1.8 million observations), additionally requiring finite ROIC values over a three-year window (t through $t + 2$). ROIC is defined as operating income at t divided by tangible fixed assets net of land at t , where t is the same period in which the investment decision captured by the capex ratio is observed.²

We classify firms into three ownership categories at each point in time, exploiting the ability to match CEO names against both shareholder lists and executive rosters:

- (i) *Solo-owner* firms: the CEO appears on the shareholder list and no other executive shares the CEO's surname;
- (ii) *Family-owner* firms: the CEO appears on the shareholder list and at least one additional executive shares the surname;
- (iii) *Non-owner* firms: the CEO does not appear on the shareholder list and no executive shares the CEO's surname.

Observations falling outside these three categories — approximately 7% of the sample — are excluded (see Figure A1 in the Online Appendix for validation of the classification criteria).

3.2. Empirical model

Our baseline specification is:

$$y_{it} = \beta \cdot \text{SalesGrowth}_{i,t-1} + \sum_g \gamma_g D_{g,it}^{\text{CEO}} + \sum_g \delta_g (\text{SalesGrowth}_{i,t-1} \times D_{g,it}^{\text{CEO}}) + \sum_h \psi_h D_{h,it}^{\text{Firm}} + \sum_h \zeta_h (\text{SalesGrowth}_{i,t-1} \times D_{h,it}^{\text{Firm}}) + \sum_k \eta_k S_{it}^k + \mathbf{X}_{it}' \phi + \mu_{im} + \lambda_{s(i)t} + \varepsilon_{it}, \quad (1)$$

where y_{it} is either the capex ratio or ROIC; $D_{g,it}^{\text{CEO}}$ and $D_{h,it}^{\text{Firm}}$ are CEO age and firm age group dummies, respectively; S_{it}^k denotes CEO succession event-study indicators, comprising dummies for all pre- and post-succession periods relative to each leadership transition, with the succession year as the reference (complementing the spell-level tenure controls by absorbing the time profile of early-tenure dynamics around each transition); $\mathbf{X}_{it}$ is a vector of controls (log employees, EBITDA ratio, leverage ratio, number of partner banks, and CEO tenure group indicators and their interactions with lagged sales growth); the ROIC specification additionally includes the within-firm normalized log of tangible fixed assets net of land to absorb the mechanical negative relationship between capital stock size and the dependent variable; μ_{im} denotes firm-by-CEO fixed effects; the scale of the panel makes these high-dimensional fixed effects feasible, and they absorb all time-invariant match-level heterogeneity, ensuring that CEO age effects are not confounded by manager–firm sorting. $\lambda_{s(i)t}$ denotes industry-by-year fixed effects. Standard errors are clustered at the firm level.³

CEO age groups are defined as ≤ 39 , 40–49, 50–59, 60–69, and 70+ (reference). Firm age groups span decades from the founding year. Two sets of coefficients are of primary interest: δ_g , which captures the differential response of investment (or ROIC) to sales growth for age group g relative to the 70+ reference group, and γ_g , which captures the level difference in the dependent variable for group g when sales growth is zero. For the ownership decomposition, we estimate the model separately for each ownership type.

¹ The TSR data has been used extensively in studies of Japanese firm dynamics; see, e.g., Ito and Miyakawa (2025) for a detailed description of the dataset and its coverage.

² Re-estimating with ROIC at $t + 1$ as the dependent variable yields qualitatively similar results for the full-sample slope differential; ownership-group-specific level effects show some time-period sensitivity. Full-sample results are reported in the Online Appendix (Figure A2).

³ We use lagged sales growth rather than Tobin's Q as a proxy for investment opportunities because our sample is dominated by unlisted firms for which market-based valuations are unavailable; sales growth captures investment fundamentals more reliably in private-firm settings (Asker et al., 2015; Biddle et al., 2009). For the same reason, we measure investment quality using ROIC rather than stock-return-based measures.

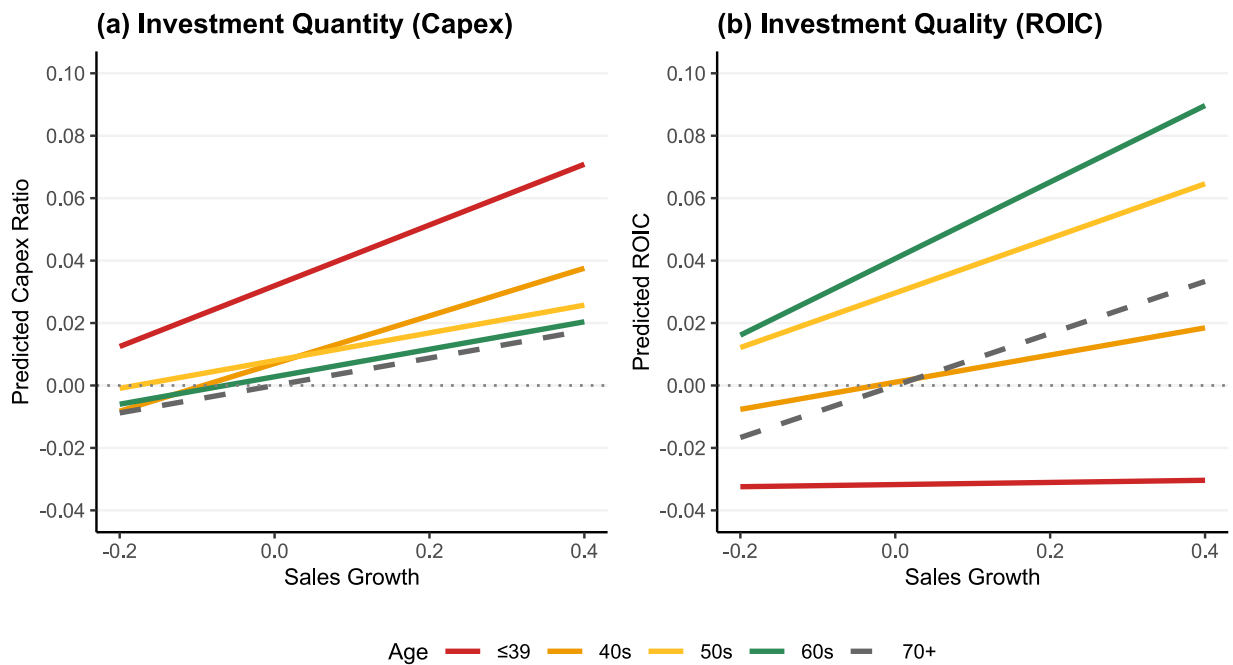


Fig. 1. CEO Age and Investment: Quantity Versus Quality. Lines show the estimated relationship between lagged sales growth and the capex ratio (left) or ROIC (right) for each CEO age group, based on the full-sample regressions in column (1) of Tables A2 and A3 (other covariates excluded). The capex ratio is the year-over-year change in tangible fixed assets net of land plus depreciation, divided by the lagged stock; ROIC is operating income divided by tangible fixed assets net of land. CEO age 70+ is the reference group.

4. Results

4.1. Quantity versus quality: full sample

Fig. 1 presents the predicted capex and ROIC responses to lagged sales growth for each CEO age group, based on the full-sample regressions reported in column (1) of Tables A2 and A3 in the Online Appendix. Differences in the slopes of the plotted lines correspond to the age-group interaction coefficients (δ), and differences in their vertical positions to the age-group base effects (γ); each visual pattern discussed below thus corresponds to a tabulated estimate. In the left panel (Quantity), CEOs aged 39 and under exhibit the steepest investment response to growth opportunities. The slope flattens progressively for older age groups. Relative to the reference group of CEOs aged 70 and over, the interaction coefficient $\delta_{\leq 39}$ is 0.053 ($p < 0.01$) and δ_{40s} is 0.032 ($p < 0.01$).

The right panel (Quality) reveals a different pattern. While the baseline ROIC response to sales growth is positive for the 70+ reference group ($\beta = 0.083$, $p < 0.10$), the interaction coefficient for the youngest group is significantly negative ($\delta_{\leq 39} = -0.080$, $p < 0.05$). The net ROIC response for young CEOs is therefore essentially flat ($0.083 - 0.080 = 0.003$), indicating that their aggressive investment fails to yield incremental returns. CEOs in their 60s are the only group with a significantly positive slope differential ($\delta_{60s} = 0.039$, $p < 0.10$).

Reinforcing this slope pattern, the ROIC base effects — which capture the average ROIC level of each age group relative to the reference when sales growth is zero, visible as the vertical separation of the lines in the right panel — show that baseline ROIC peaks among CEOs in their 50s and 60s. CEOs in their 50s and 60s exhibit significantly higher baseline ROIC ($\gamma_{50s} = 0.030$, $p < 0.05$; $\gamma_{60s} = 0.041$, $p < 0.01$), whereas the ≤ 39 group shows no ROIC advantage ($\gamma_{\leq 39} = -0.032$, $p > 0.10$).⁴ Taken together, these results confirm the hypothesized quantity–quality divergence: young CEOs respond more aggressively to growth opportunities but fail to translate that responsiveness into returns, consistent with reputation-building rather than career-concern conservatism (Prendergast and Stole, 1996; Holmström, 1999).

4.2. Heterogeneity across governance environments

Fig. 2 presents the predicted responses separately for each ownership type, revealing heterogeneity in the quantity–quality relationship; the underlying estimates are reported in columns (2)–(4) of Tables A2 and A3. We interpret these categories as distinct

⁴ The positive ROIC gradient for CEOs in their 50s and 60s is broadly consistent with recent evidence documenting an inverted U-shaped relationship between managerial age and firm sales performance, with performance peaking around age 60, using the same TSR dataset (Hoshi et al., 2025; Miyakawa et al., 2025).

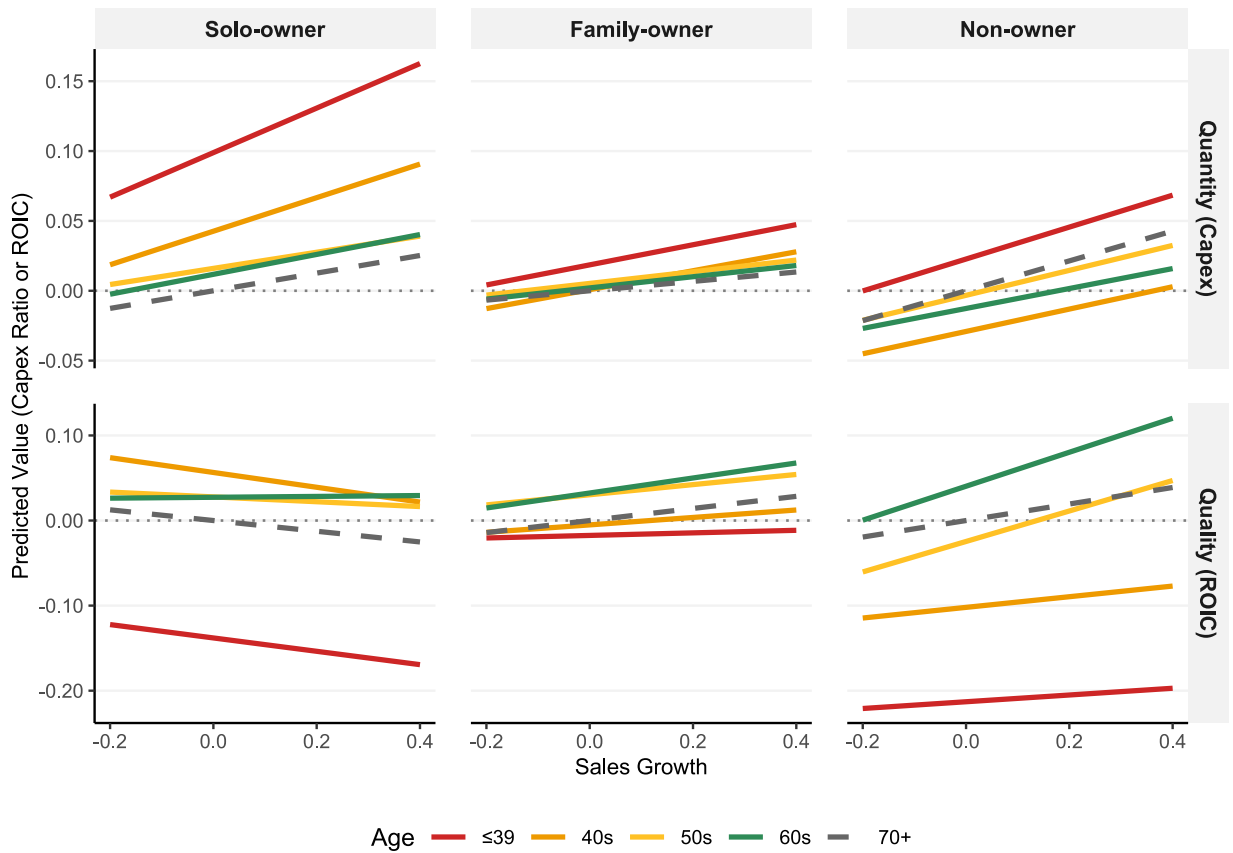


Fig. 2. CEO Age and Investment by Ownership Type. The top row displays the predicted capex ratio (Quantity), and the bottom row displays the predicted ROIC (Quality). Columns correspond to solo-owner, family-owner, and non-owner firms. Lines show the estimated relationship between lagged sales growth and capex ratio (top row) or ROIC (bottom row) for each CEO age group, based on the ownership-subsample regressions in columns (2)–(4) of Tables A2 and A3 (other covariates excluded). Variable definitions follow Figure 1. CEO age 70+ is the reference group.

governance environments rather than as causal treatments: the estimates characterize how the age-related pattern differs across environments, not the effect of ownership itself.

In solo-owner firms, the quantity–quality gap is most visible. Young solo-owner CEOs display the highest capex sensitivity across all ownership types ($\delta_{\leq 39} = 0.096, p < 0.10$; upper-left panel of Fig. 2). However, their ROIC response to growth opportunities is statistically indistinguishable from zero (lower-left panel): neither the baseline slope nor the interaction term for the ≤ 39 group is significant. This pattern is consistent with the free-cash-flow agency costs of unchecked managerial discretion (Jensen, 1986; Stulz, 1990).

In family-owner firms, the pattern is distinct. Young CEOs (aged 39 and under) and those in their 40s retain moderate but statistically significant investment sensitivity compared to the 70+ reference group ($\delta_{\leq 39} = 0.038, p < 0.10$; $\delta_{40s} = 0.034, p < 0.01$; upper-middle panel). While this sensitivity is less than half of that observed in solo-owner firms for the youngest group, it confirms that youthful dynamism persists even under family oversight. Crucially, the investment quality penalty is mitigated: the ROIC base effect for the youngest group is small and statistically insignificant ($\gamma_{\leq 39} = -0.018, p > 0.10$; lower-middle panel), as is the ROIC slope differential. This evidence is consistent with kinship-based monitoring curbing the most speculative investments while preserving managerial initiative.

Among non-owner firms, the pattern is driven by an efficiency deficit rather than excess investment. Young professional managers exhibit no significant differential in investment sensitivity ($\delta_{\leq 39} = 0.007, p > 0.10$; upper-right panel), but suffer a large baseline ROIC penalty ($\gamma_{\leq 39} = -0.213, p < 0.05$; lower-right panel). That is, young non-owner CEOs invest at a similar rate to older counterparts yet operate at lower efficiency. This is consistent with career-concerns models in which equity-less managers prioritize visible outcomes over long-run returns (Gibbons and Murphy, 1992).

4.3. Robustness

Several features of our design address identification concerns. Firm-by-CEO fixed effects absorb all time-invariant match-level heterogeneity; the identifying variation comes from within-match changes in growth opportunities. Every regression controls for

CEO tenure flexibly via one-year tenure indicators interacted with lagged sales growth.⁵ Firm age controls and CEO succession indicators are included throughout. Table A4 confirms stability of the key age coefficients under progressively stricter fixed effects, including with flexible tenure interactions.

The remaining analyses address endogeneity more directly. Tenure-stage subsamples show the young-CEO investment-sensitivity ordering is already visible among newly appointed CEOs and does not grow with tenure (Online Appendix OA.2). Clean succession events confirm causal direction: pre-succession paths show no anticipatory trend, and the quality differential emerges post-succession, consistent with the baseline pattern (OA.3). A doubly robust estimator confirms the pattern is robust to selection on observables (OA.4). Ownership type is stable within firm–CEO spells, validating the subsample approach (OA.5).

5. Conclusion

Estimating the quantity and quality of investment within a single unified framework, this paper documents a fundamental asymmetry in the behavior of young versus old CEOs: young CEOs invest more in response to growth opportunities, yet the quality of their investments — measured by ROIC — is systematically lower. Because both margins are estimated under the same fixed-effects structure and against the same growth signal, this divergence emerges within a single design rather than from a juxtaposition of estimates across disparate settings, and it implies that youth-driven investment activity is not unambiguously beneficial.

Our ownership decomposition reveals that the severity and form of this trade-off vary across governance environments. The quantity–quality trade-off is most severe in solo-owner firms, consistent with unchecked managerial discretion. In family-owner firms, the efficiency penalty is largely absent, consistent with kinship-based internal monitoring preserving investment quality across CEO generations. In non-owner firms, young professional managers invest at comparable rates yet exhibit an efficiency deficit, consistent with career concerns rather than unchecked discretion.

Two implications follow for the corporate finance literature. For work on capital allocation efficiency (Richardson, 2006), investment responsiveness alone is an incomplete metric: a quantity-only design would record young CEOs as the most responsive to growth opportunities while missing that this responsiveness yields no incremental return. For work on managerial incentives, which margin early-career reputational pressure distorts varies with the governance environment: the quantity margin where discretion is unchecked (Prendergast and Stole, 1996), the quality margin where equity incentives are absent (Holmström, 1999; Gibbons and Murphy, 1992). Tests of career-concern models may thus need to condition on the ownership and monitoring environment in which managers operate.

Limitations include the use of ROIC as a coarse quality measure and the inability to observe project-level returns. Future work could extend this framework to R&D efficiency and board structure heterogeneity.

CRedit authorship contribution statement

Yojiro Ito: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daisuke Miyakawa:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) in order to assist with refining the manuscript text and checking the consistency of arguments. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.frl.2026.110499>.

Data availability

The data that has been used is confidential.

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⁵ Tenure is observed only for spells whose start date falls within the sample window; otherwise it is left-censored. Left-censored spells are retained via a *tenure_Missing* indicator and its interaction with sales growth, preserving the full estimation sample.

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